Get started with lakehouses in Microsoft Fabric

1. Core Capabilities of Microsoft Fabric

- **Unified Analytics Platform**: Fabric is a software-as-a-service (SaaS) platform that integrates data engineering, data integration, warehousing, real-time intelligence, data science, and business intelligence into a single environment.
- **OneLake Storage**: This is the centralized data storage architecture that unifies data across regions and clouds into a single logical lake. All compute engines store their data in OneLake, typically in the open **Delta-Parquet** format for tabular data.
- **Shortcuts**: This feature allows users to virtualize and access data from external sources (like Amazon S3 or Google) within OneLake without moving or duplicating the original data.
- **Copilot Integration**: A generative AI assistant is integrated across all workloads to enhance productivity through intelligent code completion, natural language-to-SQL conversion, and automated insights.

2. The Role of the Lakehouse

- **Hybrid Foundation**: A lakehouse combines the flexible, scalable storage of a data lake with the querying and analytical capabilities of a relational data warehouse.
- **Key Features**:
    - **Schema-on-read**: Users define the schema as needed rather than adhering to a predefined one.
    - **ACID Transactions**: Supported through Delta Lake formatted tables to ensure data consistency and integrity.
    - **Multiple Access Modes**: When a lakehouse is created, it automatically provides a **SQL analytics endpoint** for read-only SQL queries and a **Semantic model** for Power BI reporting.

3. Team Collaboration and Workloads

- **Removing Silos**: Fabric allows different data professionals (engineers, analysts, and scientists) to collaborate effectively without duplicating efforts or systems.
- **Workload-Specific Tools**:
    - **Data Engineering**: Ingests and transforms data using Spark notebooks and pipelines.
    - **Data Science**: Built-in support for Python and Spark to build and test machine learning models.
    - **Data Analysts**: Can connect directly to OneLake using **Direct Lake mode**, reducing the need for downstream transformations.
    - **Citizen Developers**: Can discover curated data via the **OneLake catalog** and use Power BI templates for quick reporting.

4. Administration and Security

- **Enabling the Service**: Fabric must be enabled in the **Admin portal** under tenant settings by a Fabric, Power Platform, or Microsoft 365 admin.
- **Workspaces**: These serve as logical containers to organize assets and provide clear separation of resources and security.
- **Governance**: Centralized administration allows for managing permissions, monitoring usage, and using the OneLake catalog to track metadata and sensitivity labels.

Questions

1. What is a key benefit of using Microsoft Fabric in data projects?
A. It allows data professionals to work independently, without collaboration.
B. It requires duplicating data across systems to ensure availability.
C. It provides a single, integrated environment for collaboration on data projects.
D. It eliminates the need for any cloud-based storage or internet connection.

2. What is the default storage format for Fabric's OneLake?
A. Delta-Parquet
B. JSON
C. CSV
D. XML

3. Which Fabric experience is used to move and transform data?
A. Data Science
B. Data Warehousing
C. Power BI
D. Data Factory

4. Which feature of Microsoft Fabric lakehouses supports machine learning and predictive modeling analytics?
A. Schema-on-read format
B. Spark and SQL engines
C. Item-level sharing
D. Sensitivity labels

5. When designing an ETL process for a Microsoft Fabric lakehouse, which tool can be used to visually represent transformations without traditional programming?
A. Notebooks
B. SQL analytics endpoint
C. Dataflows Gen2
D. OneLake shortcuts

6. Your organization plans to train machine learning models using data from a Microsoft Fabric lakehouse. Which tool is most appropriate for performing this task?
A. SQL analytics endpoint
B. Notebooks
C. Data Factory pipelines
D. OneLake catalog

7. What is a significant benefit of schema-on-read in Microsoft Fabric lakehouses compared to predefined schemas?
A. It requires significantly less storage space.
B. It ensures faster data retrieval than predefined schemas.
C. It allows for more flexible data ingestion and transformation.
D. It eliminates the need for any data security or governance.

8. How does a Microsoft Fabric lakehouse differ from a typical data lake in terms of data processing?
A. Lakehouses automatically transform all data into structured formats.
B. Data lakes inherently support real-time analytics without additional tools.
C. Lakehouses integrate SQL and Spark engines for advanced data processing.
D. Data lakes provide built-in Power BI reporting by default.

9. Your organization wants to combine structured transaction data with social media feeds for analysis. Why is a Microsoft Fabric lakehouse suitable?
A. Traditional data warehouses do not support SQL queries.
B. Lakehouses can store all data formats and support SQL-based analytics.
C. Lakehouses require less storage space than traditional data warehouses.
D. Traditional warehouses cannot connect to internet-based data sources.

10. What serves as the centralized, "logical lake" architecture for all data in Microsoft Fabric?
A. Azure SQL Database
B. OneLake
C. Data Factory
D. Power BI Workspace

11. Which feature allows you to virtualize data from external sources like Amazon S3 without moving or duplicating it?
A. Mirroring
B. Pipelines
C. Shortcuts
D. Notebooks

12. Which administrative role is authorized to manage Fabric settings and configurations in the Admin portal?
A. Data Analyst
B. Fabric admin
C. Data Scientist
D. Contributor

13. In Microsoft Fabric, what are the logical containers used to organize data, reports, and other assets?
A. Shortcuts
B. Domains
C. Workspaces
D. Engines

14. Which Fabric component is used to process, monitor, and analyze streaming data in real-time?
A. Real-Time Intelligence
B. Data Warehouse
C. Data Engineering
D. Industry Solutions

15. What is the role of the "SQL analytics endpoint" in a lakehouse?
A. It allows users to write and execute Spark notebooks.
B. It provides read-only access to query lakehouse tables with SQL.
C. It is the primary tool for ingesting unstructured image files.
D. It replaces the need for OneLake storage entirely.

16. Which workspace role is most appropriate for a collaborator who needs to create and manage items?
A. Viewer
B. Contributor
C. Guest
D. Security Admin

17. How does Copilot in Fabric assist a Data Warehouse user?
A. By automatically deleting old records to save space.
B. By converting natural language into SQL queries.
C. By designing physical server hardware layouts.
D. By managing the company's financial budget.

18. Which tool is best for orchestrating different ETL activities within Microsoft Fabric?
A. Data Factory pipelines
B. Power BI reports
C. OneLake catalog
D. SQL analytics endpoint

19. What does the "Direct Lake" mode allow Data Analysts to do?
A. Export data to local Excel files for analysis.
B. Connect directly to OneLake, reducing the need for downstream transformations.
C. Use Spark to write complex Python scripts for machine learning.
D. Access the internal source code of the Fabric platform.

Answer Key

1. C
2. A
3. D
4. B
5. C
6. B
7. C
8. C
9. B
10. B
11. C
12. B
13. C
14. A
15. B
16. B
17. B
18. A
19. B